

Gluconeogenesis

Adult human consumes 160g of glucose per day, most of which is used by brain (120g per day) and skeletal muscle.

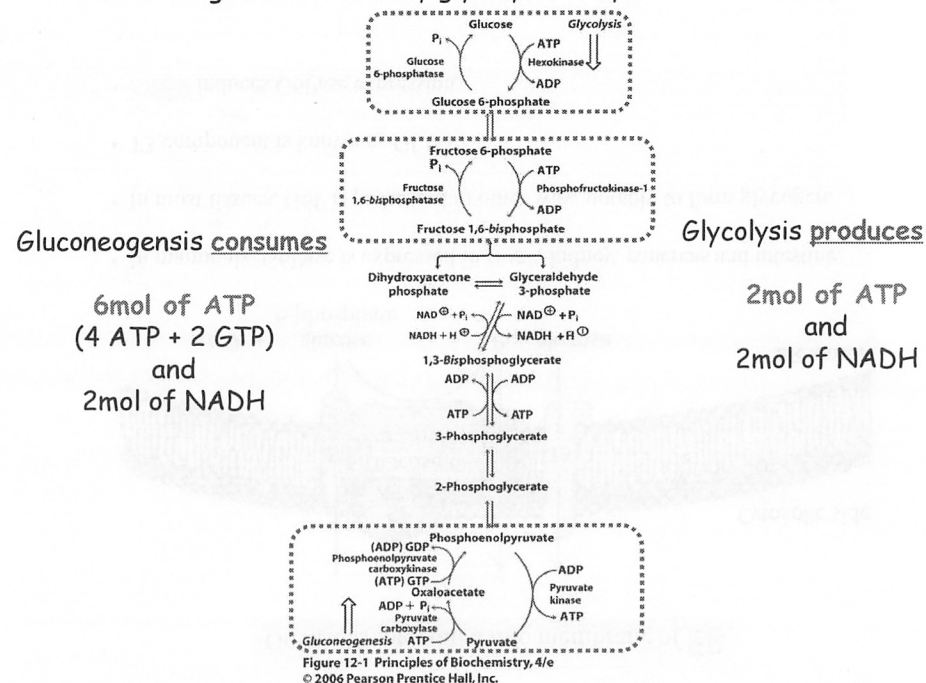
The amounts of glucose in serum are about 20g, and those from glycogen are about 190g. Therefore, direct glucose reserves are sufficient to meet glucose needs for only about a day.

Role for gluconeogenesis:

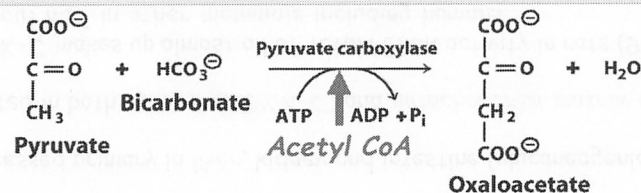
1. Provision of glucose as an **energy source**, essential in brain, kidney medulla, lens, cornea and testis. ➡ Maintenance of blood glucose level
2. Control of **acid-base balance**. Re-synthesis of glucose from lactate is a major route of lactate disposal.
3. Maintenance of **amino acids balance**.
4. Provision of **biosynthetic precursors** such as glycolipids, glycoproteins.

Gluconeogenesis occurs preferentially in **liver** and cortex of **kidney**

Gluconeogenesis uses many glycolytic enzymes in the reverse direction

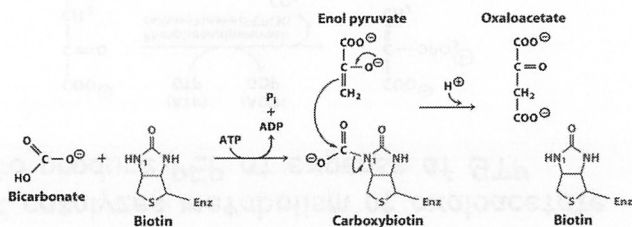


Pyruvate is carboxylated to Oxaloacetate in Mitochondria

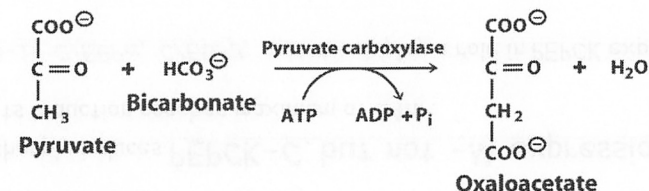


Unnumbered figure pg 360a Principles of Biochemistry, 4/e
© 2006 Pearson Prentice Hall, Inc.

- Pyruvate in cytosol is transported into mitochondria by Mitochondrial Pyruvate Carrier (MPC)
- In this reaction, **biotin** is required as a co-factor and serves as a **carrier of activated CO₂**.
- **Acetyl CoA**, produced by **FA oxidation**, is an allosteric effector and activates the enzyme. Higher concentration of acetyl CoA implies insufficient metabolism in TCA cycle.



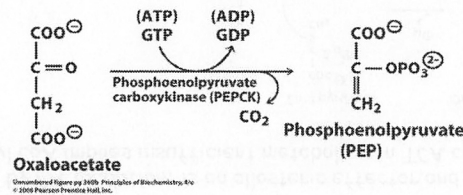
Pyruvate is carboxylated to Oxaloacetate in Mitochondria



Unnumbered figure pg 360a Principles of Biochemistry, 4/e
© 2006 Pearson Prentice Hall, Inc.

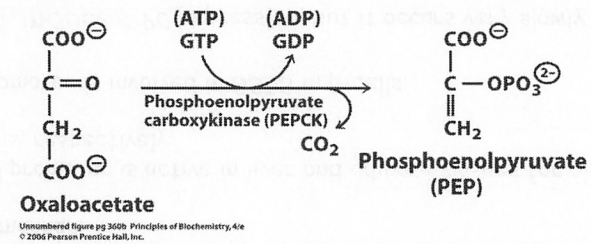
- Alternative promoter (two in rodents and three in human) is used in a tissue-dependent manner.
 - Proximal promoter is active in liver and adipose tissues for gluconeogenesis and lipogenesis, respectively.
 - Distal promoter is involved in GSIS in β-cells.
- **Glucagon induces PC expression**, but it occurs very slowly **at 72h after starvation**.

PEPCK catalyzes metabolism of oxaloacetate to produce PEP at expense of GTP



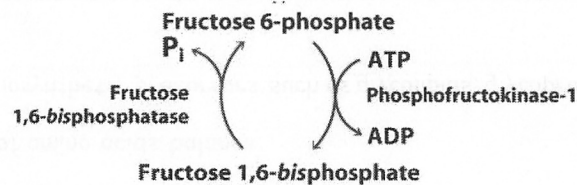
- In mammals, PEPCK requires **GTP** as a phosphate donor.
- PEPCK enzyme activity determines rate of gluconeogenesis.
- PEPCK is expressed primarily in **liver**, kidney and intestine (gluconeogenic tissues).
- PEPCK is located in both **cytosol (PEPCK-C)** and **mitochondrial matrix (PEPCK-M)**.
- In liver, **PEPCK-C** makes up almost all of total PEPCK activity in rats (95%) and mice (99%), but about half in other mammals including humans.

PEPCK catalyzes metabolism of oxaloacetate to produce PEP at expense of GTP



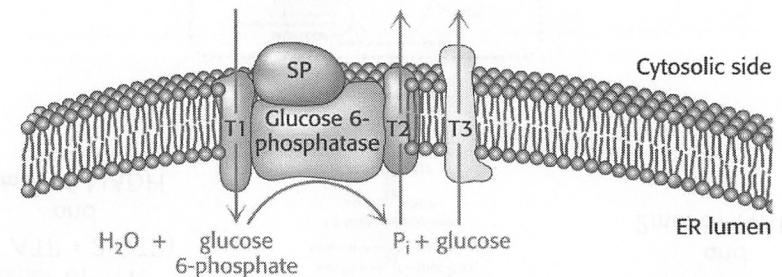
- **PEPCK activity** is controlled exclusively at a **transcriptional level** by changing its expression, but not by allosteric regulator and post-translational modification.
- **Glucagon** sharply induces **PEPCK-C**, but not **-M** expression in a few **hours** and its induction reaches maximum at 24hr.
- **CREB**, **FOXO-1**, **C/EBPα**, **C/EBPβ**, and **AP-1** play a role in PEPCK expression.

FBPase



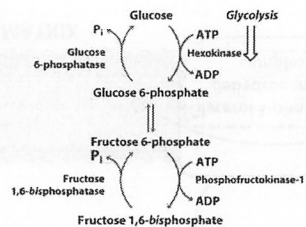
- Two isoforms, **FBPase-1** and **-2** is expressed in liver and muscle, respectively.
- Both isoforms are inactivated by **Fructose-2, 6-Bisphosphate**.
- Distinct from PEPCK and G6Pase, **FBPase is not induced by cAMP-glucagon**

G6Pase is integrated into membrane of ER



- In mammals, G6Pase is expressed in **liver**, kidney, pancreas and intestine.
- In most tissues, G6P is processed in other way, notably to form glycogen.
- T3 component is known as GLUT7.
- **CREB** induces G6Pase expression.

Major precursor
for gluconeogenesis



★ **Glycerol** → Dihydroxyacetone phosphate ⇌ Glyceraldehyde 3-phosphate ← **Fructose**

Glycerol from adipose tissues
(TG), where lacks glycerol kinase !!
(check later)

Lactate from skeletal muscle, RBCs, skin and brain (check in next slide)

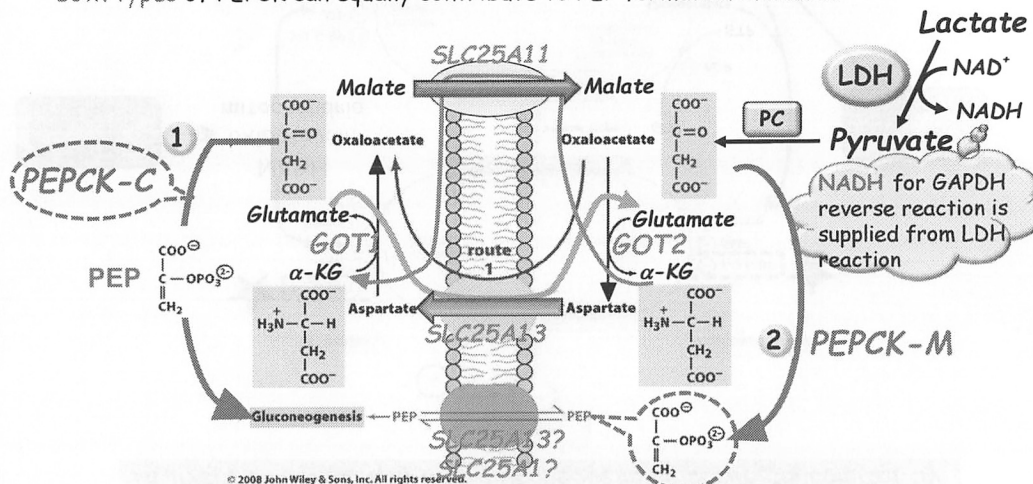
★ Amino acids
Propionate

Lactate ★
Amino acids ★

Figure 12-1 Principles of Biochemistry, 4/e
© 2006 Pearson Prentice Hall, Inc.

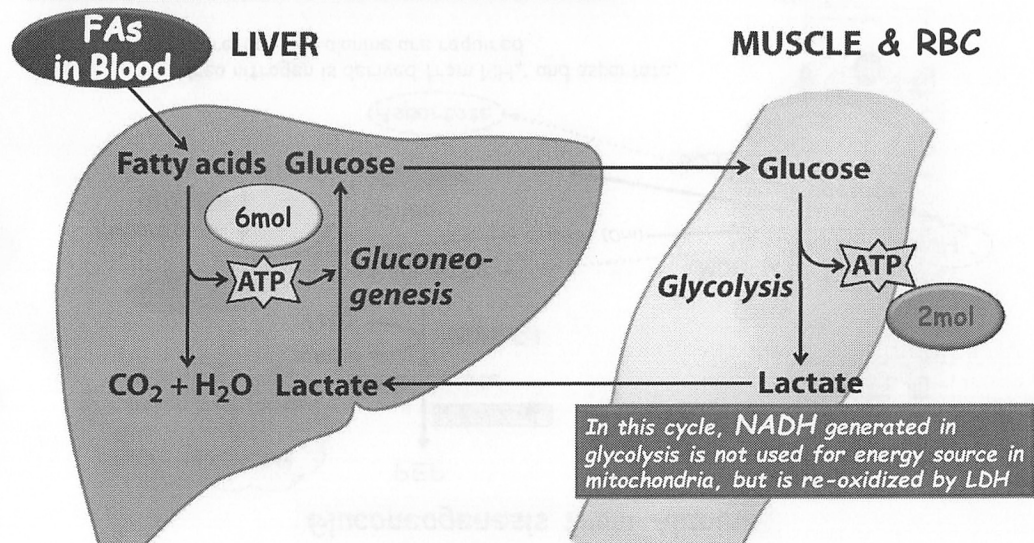
Gluconeogenesis from Lactate

- Mitochondrial oxaloacetate derived from lactate is metabolized to either aspartate or PEP for transport to cytosol.
- ***In starvation***, highly expressed PEPCK-C plays a role to produce PEP in mice, while both types of PEPCK can equally contribute to PEP formation in human.

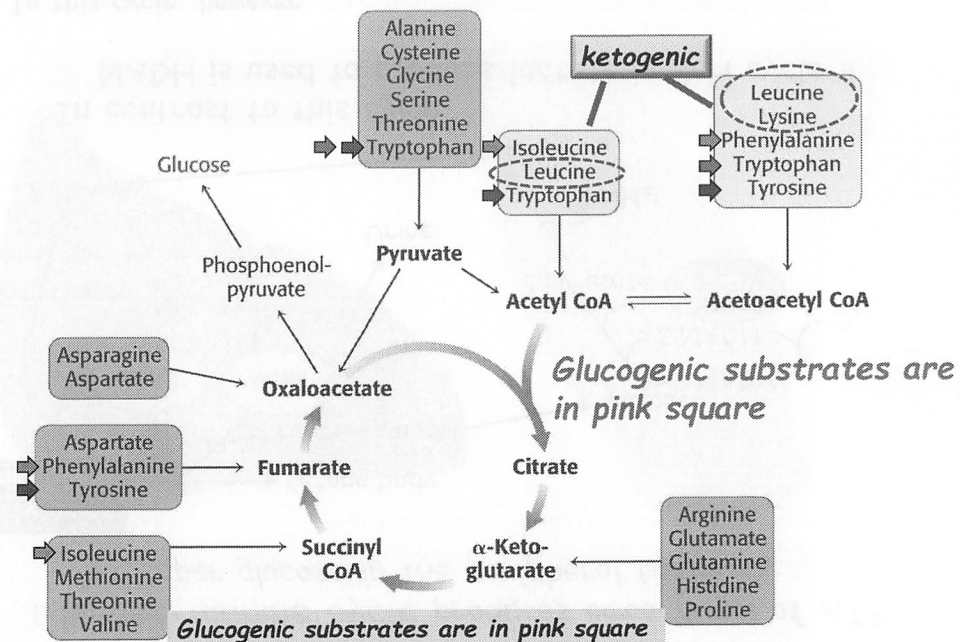


Cori cycle produces only 2 mol of ATP per glucose in the peripheral tissues

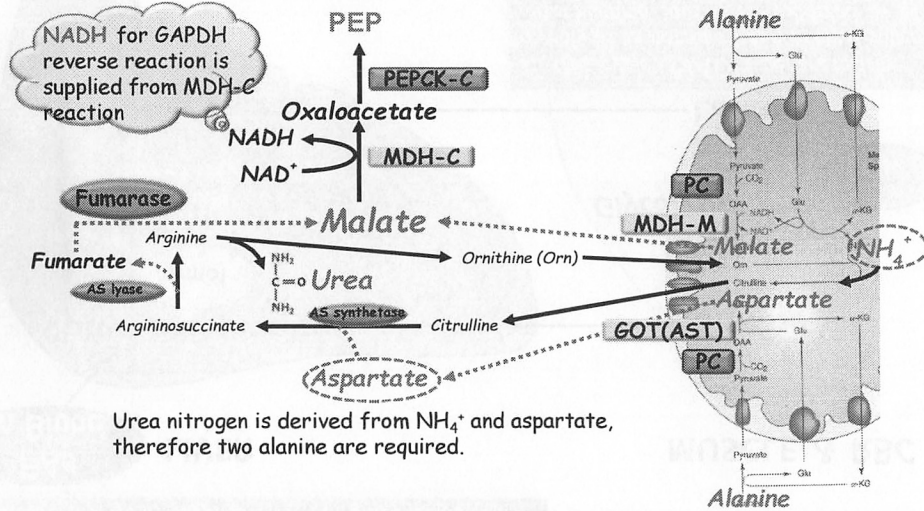
In this cycle, oxidation of FA derived from adipose tissue (lipolysis) supplies ATP for gluconeogenesis



Glucose is synthesized from most amino acids



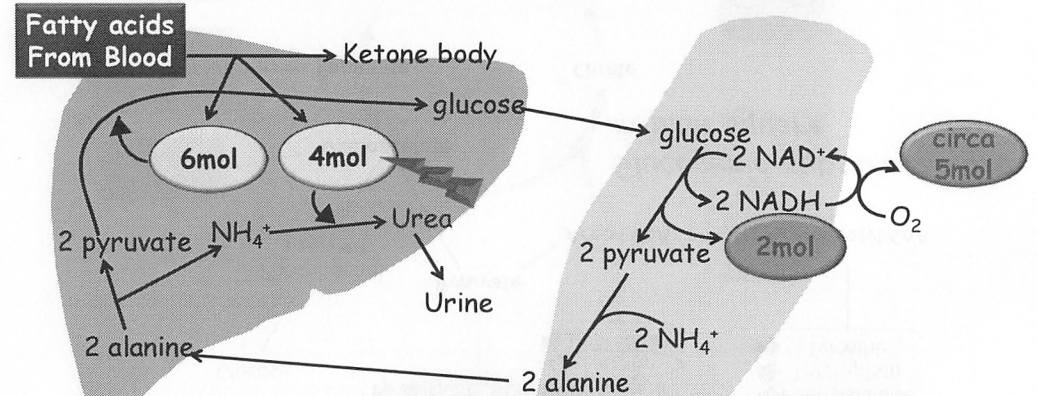
Gluconeogenesis from Alanine



Gluconeogenesis and Urea synthesis should work cooperatively

Liver disposes nitrogen of alanine as urea

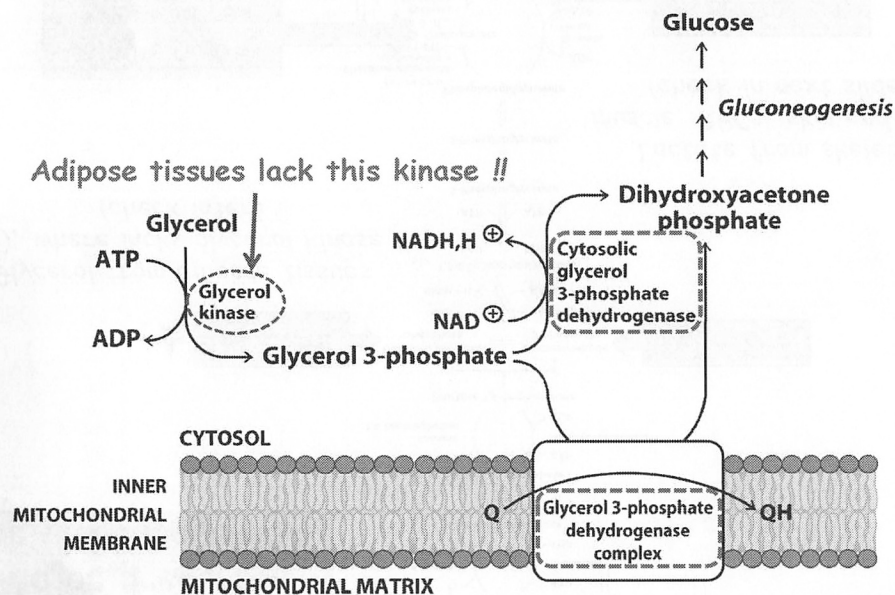
Glucone-alanine cycle produces circa 7 mol of ATP per glucose in the peripheral tissues



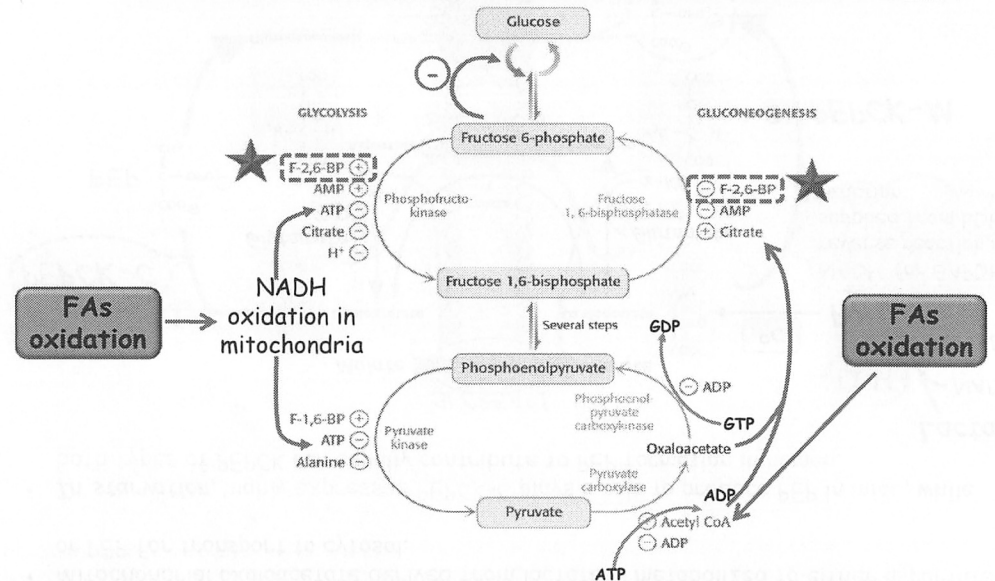
In contrast to this cycle, NADH is used to produce lactate in Cori cycle !!

In this cycle, however, liver consumes more ATP to dispose nitrogen from amino acids!!

Glycerol is metabolized to glucose in the liver

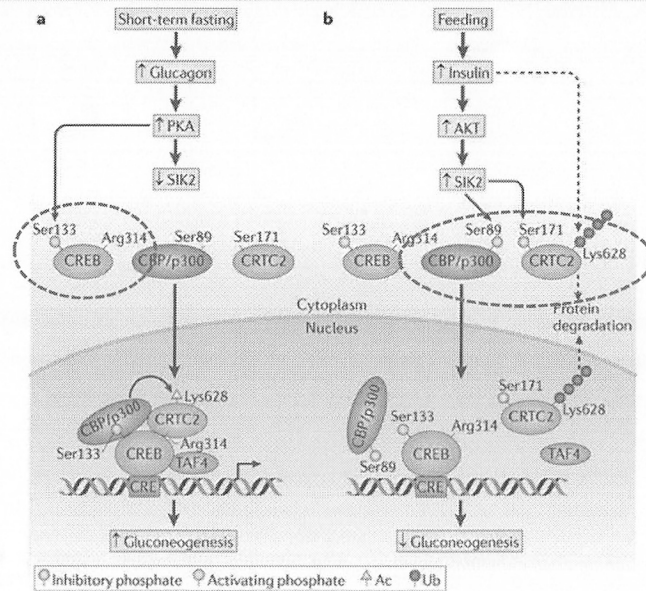


In liver, FA oxidation and gluconeogenesis occur concomitantly



FA oxidation provides not only ATP for gluconeogenesis, but also positive and negative allosteric regulators for gluconeogenesis and glycolysis, respectively

Glucagon and Insulin signaling pathway to gluconeogenesis



Nature Reviews | Molecular Cell Biology

CREB: cAMP response element binding protein CRTC2: CREB Regulated Transcription Coactivator 2

